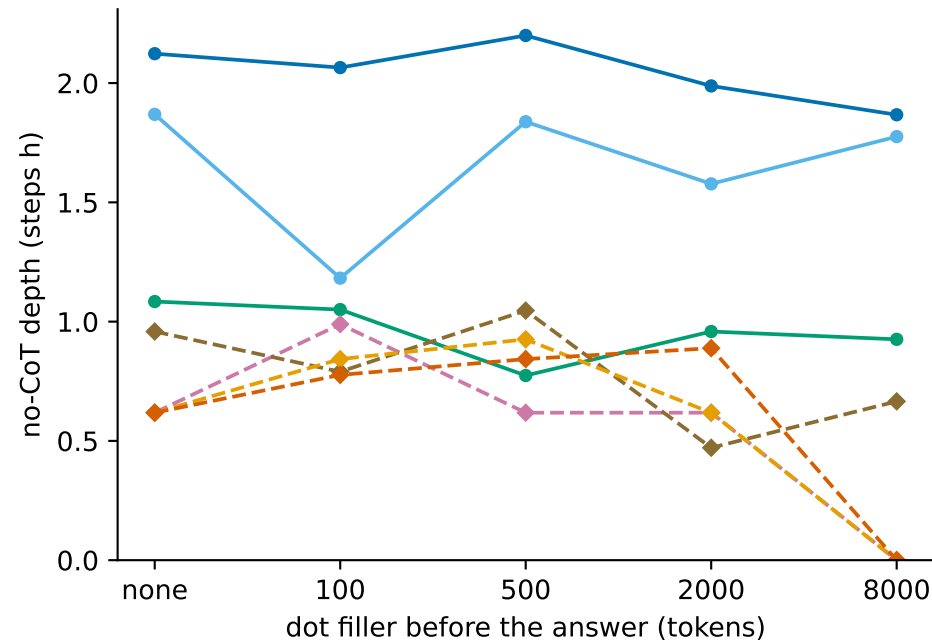


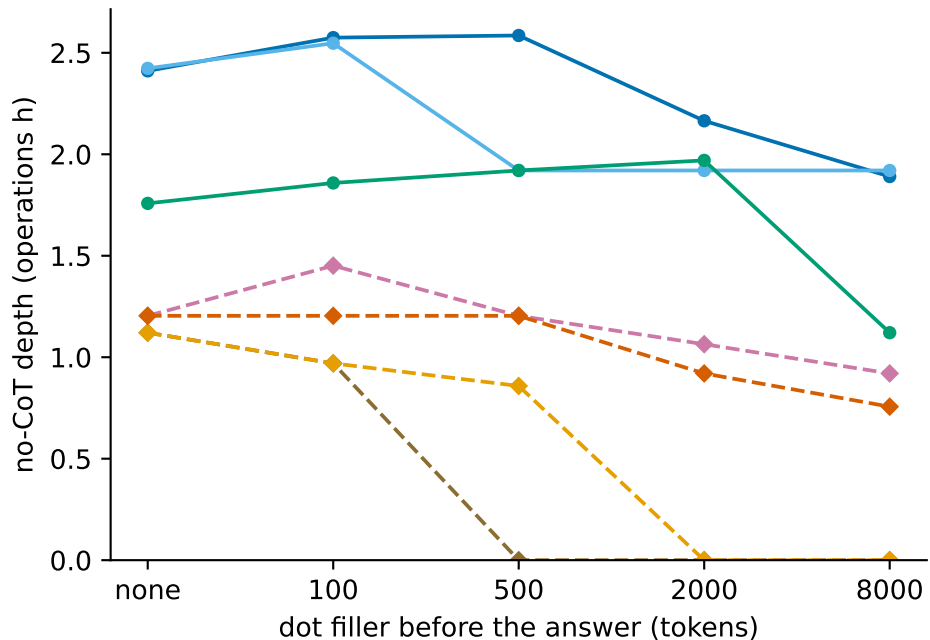
Dot filler buys no extra depth: no-CoT depth stays flat or falls as the filler grows

- Gemma-4 31B (standard)
- Qwen3 32B (standard)
- Nanbeige4.2 3B (looped)
- Ouro 2.6B (looped)
- Qwen3.8 27B (standard)
- IQuest 40B Loop (looped)
- Ouro 1.4B (looped)

Chain



Arithmetic



Solid: standard; dashed: looped. Depth fitted as in figure 1 (0 = at chance even for one step); dev items only. 8000 dots push the small looped models to chance.